

ARUN MALLIKARJUNA

AI Platform & Full-Stack Engineer | LLM Integration · Agentic Apps · System Design

arunmallikarjun005@gmail.com · +91 74835 73363 · Bengaluru, India · LinkedIn · Portfolio

PROFESSIONAL SUMMARY

Full-stack engineer with 4+ years building and shipping production systems end to end — frontend, backend, and infrastructure — and 2+ years integrating LLM APIs (Gemini, Claude) and building core product features around them. Hands-on building agentic apps — custom AI agents with tool-calling, multi-step orchestration, subagents, system-prompt design, RAG pipelines, and evals — backed by strong system-design judgment and a product-driven mindset that ties code to the real customer problem. Comfortable owning the full lifecycle solo or in a small team — debugging and fixing failing code, reviewing implementations against the PRD, and running thorough QA (SIT, smoke, UAT, VAPT) before release.

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Tecnotree Convergence

Jan 2024 – Present · Bengaluru, India

- Owned architecture and technical decisions for a rule-based Promotion Engine — the team's go-to for design choices — letting business teams launch telecom campaigns without redeployment and removing engineering from the campaign-launch loop.
- Took the lead when systems failed: independently debugged and fixed production bottlenecks across distributed microservices, sustaining 99.5% availability under high transaction volume.
- Reviewed code and architecture against the PRD and ran thorough QA — SIT, smoke testing, UAT, and coordinated VAPT — to ship reliable releases.
- Made pragmatic performance and scope trade-offs without over-engineering — improved frontend performance ~30% via optimized React rendering, and tuned rule evaluation to cut per-request latency.
- Integrated third-party services and contributed to CI/CD and deployment workflows to ship reliably across the stack.
- Built reporting dashboards and analytics that surfaced campaign performance, aligning engineering output to clear business outcomes.

Software Engineer — Tecnotree Convergence

Oct 2021 – Dec 2023 · Bengaluru, India

- Built scalable REST APIs in Node.js for high-volume transaction processing, optimizing for low latency under concurrent load with async/await and event-driven patterns.
- Contributed to the microservices architecture, improving end-to-end throughput and system performance.
- Developed configurable campaign workflows so non-technical business teams could launch promotions independently — aligning the code to a real customer and business need.
- Optimized promotion rule-evaluation logic to reduce per-request latency and improve processing efficiency.

PROJECTS

Tailorwright — Multi-Surface AI Agent Platform

Next.js · TypeScript · Node.js · NestJS · Claude API

- Built a full-stack TypeScript system with four front-ends — a web app, a Claude Code skill, a custom subagent, and a CLI — from one shared agent core, spanning frontend, backend, and tooling.

- Designed the agent architecture: authored the system prompts, wired multi-step tool-calling and a delegated subagent, and built a planner/executor loop that reads inputs, decides actions, and self-corrects on failure.
- Enforced a strict no-fabrication guardrail and structured-output schemas so the agent's responses stay grounded and machine-validatable, with retries on schema mismatch.
- Built a deterministic, evaluation-ready scoring core and ran evals to measure agent output quality across cases, iterating on prompts to raise pass rates.
- Instrumented the agents for end-to-end debuggability and hardened the web app against SSRF and malicious input.

Free Resume Pro — AI SaaS Platform

Next.js · React.js · Node.js · Gemini API · Vercel

- Designed, built, and launched a full-stack AI SaaS product end to end as the sole developer — frontend, backend, and deployment — owning the entire lifecycle on a one-person team.
- Integrated the Gemini API with structured prompt engineering and a retrieval-augmented (RAG) flow over embeddings and vector retrieval for ATS scoring, job-description matching, and AI résumé rewriting.
- Architected a privacy-first system with zero user-data storage, shipped 12+ customizable templates with inline editing, and scaled it on Vercel — making pragmatic trade-offs to ship fast without over-engineering.

Promotion Manager System — Internal Platform

Node.js · React.js · Microservices · REST APIs

- Designed and built a scalable promotion-management system for telecom campaigns on a microservices backend.
- Built a rule-based engine that evaluates eligibility and applies promotions in real time, enabling non-technical teams to configure campaigns without code deployments.
- Optimized rule processing to improve throughput and reduce evaluation latency across high-volume traffic.

TECHNICAL SKILLS

Languages

JavaScript (ES6+), TypeScript, Python

AI & LLMs

LLM API integration (Gemini, Claude), AI Agents / Agentic Apps, Tool-Calling & Multi-Step Orchestration, Subagents, Prompt & System-Prompt Engineering, RAG pipelines, Embeddings & Vector Retrieval, Structured Outputs, Evals

Backend

Node.js, NestJS, Express.js, GraphQL, Apollo, Django, Flask, REST API design, Async/Await · Promises · Event Loop

Frontend

React.js, Next.js, State Management, Routing, HTML5, CSS3, Tailwind CSS

Databases

PostgreSQL (SQL), MongoDB · MongoDB Atlas (NoSQL), Schema Design

Architecture & Infra

Microservices, System Design, Design Patterns, Reliability & Scaling, Serverless, CI/CD, Vercel, Railway

Auth & Security

Firebase Authentication, JWT, Clerk, Authentication & Authorization, SSRF / Input Hardening

QA & Testing

SIT, Smoke Testing, UAT, VAPT

Tools & Methods

Git, Jenkins, Stripe, Agile, Scrum

EDUCATION

Bachelor of Engineering — Computer Science

Maharaja Institute of Technology, Mysuru · Graduated 2021